

# Rectal adenocarcinoma

Neoadjuvant chemoradiation is the hallmark deviation from the colon strategy — metastatic regimens are otherwise the same. Staging, prognosis and most systemic therapy defer to the colon page

Stanford Oncology Handbook 2025, Rectal Cancer (Tam AY, Shaheen S) · NCCN · items marked † are outside the handbook chapter

## WORKUP

- Colonoscopy + biopsy with **MMR / MSI testing** and pathology review
- MRI pelvis (3T)** — the rectal-specific study; rectal ultrasound if MRI contraindicated, inconclusive, or for superficial lesions
- CT chest/abdomen/pelvis · FDG-PET if potentially curable M1
- Labs: CBC, CMP, **CEA**
- PIK3CA testing in stage II–III** — if mutated, aspirin 100–162 mg daily × 3 years post-op
- If metastatic: KRAS, BRAF, HER2, POLE/POLD1, RET, NTRK
- Genetics referral for all rectal cancer **<50 y**, or a somatic finding with germline implications

## RISK DEFINITIONS

|                                            |                                                                                                                                             |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>High-risk rectal</b>                    | cT4a or cT4b · <b>extramural vascular invasion</b> · cN2 · <b>involved mesorectal fascia</b> · enlarged lateral nodes considered metastatic |
| <b>High-risk path after local excision</b> | positive margins · lymphovascular invasion · poorly differentiated · submucosal invasion to the lower third                                 |
| <b>T4 (rectal nuance)</b>                  | direct extension through serosa into prostate, seminal vesicles, cervix or vagina                                                           |

TNM is otherwise identical to colon — see the colon one-pager for stage groups and for all metastatic systemic therapy.

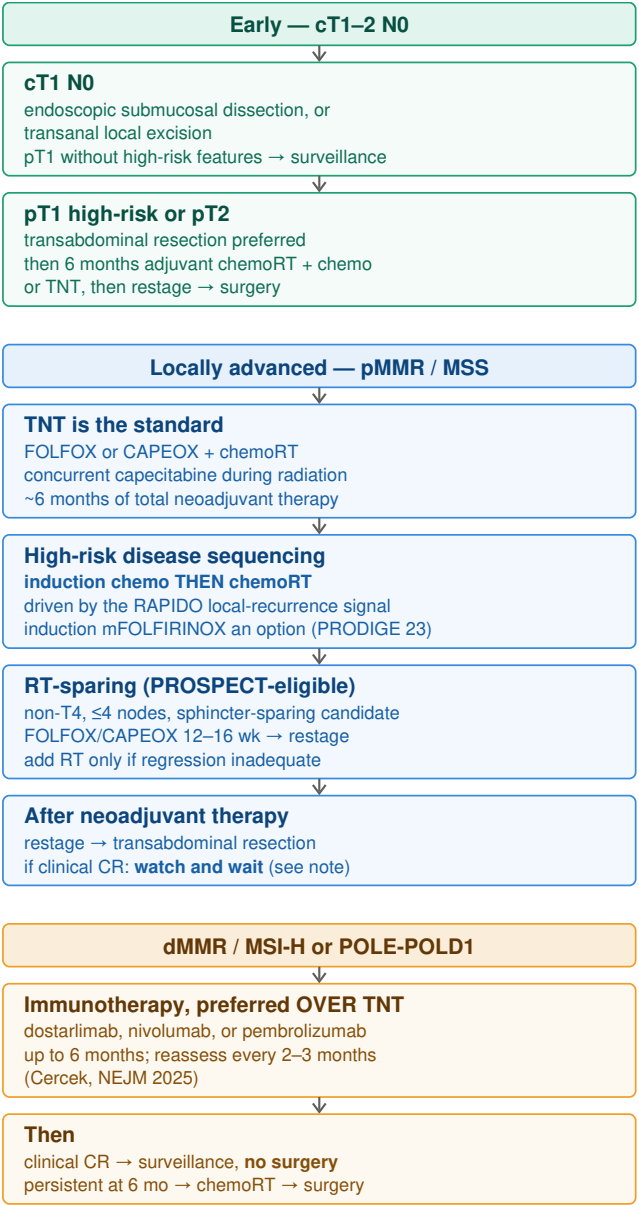
## KEY TRIALS

|                           |                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PROSPECT</b>           | RT-sparing: FOLFOX/CAPEOX × 12–16 wk with radiation reserved for inadequate regression                                                           |
| <b>RAPIDO</b>             | short-course RT + chemo TNT: pCR <b>28% vs 14%</b> ; lower distant relapse but <b>possible higher local recurrence</b> (Dijkstra, Ann Surg 2023) |
| <b>PRODIGE 23</b>         | induction mFOLFIRINOX: pCR <b>27.8% vs 12.1%</b> (Conroy)                                                                                        |
| <b>OPRA</b>               | watch-and-wait after clinical CR to TNT preserves the organ without compromising survival                                                        |
| <b>Cercek (NEJM 2025)</b> | dostarlimab in <b>dMMR</b> rectal cancer — basis for immunotherapy over TNT                                                                      |
| <b>SHORTFOX</b>           | ongoing — RT omission and course length                                                                                                          |

**Watch-and-wait requires a complete clinical response on all three of DRE, rectal MRI, and direct endoscopic evaluation** — and only in centers with experienced multidisciplinary teams, after careful discussion with the patient.

**RAPIDO caveat:** short-course RT lowered distant relapse but may raise local recurrence, so high-risk patients get TNT as induction chemo followed by chemoradiation. Short-course candidacy is a multidisciplinary decision weighing downstaging against long-term toxicity.

## TREATMENT PATHWAY



## REGIMENS & DOSING

| Regimen                              | Dosing                                                    | Schedule           |
|--------------------------------------|-----------------------------------------------------------|--------------------|
| <b>ChemoRT (infusional 5-FU)</b>     | 5-FU <b>225 mg/m²</b> IV over 24 h, 5 days per week       | with RT × 5 wk     |
| <b>ChemoRT (capecitabine)</b>        | capecitabine <b>825 mg/m² PO BID</b> on days of radiation | concurrent with RT |
| <b>Long-course radiation</b>         | 5–6 weeks                                                 | —                  |
| <b>Short-course radiation</b>        | 5 days                                                    | —                  |
| <b>Dostarlimab</b>                   | 500 mg IV × 4 doses, then 1000 mg IV                      | q3wk → q6wk        |
| <b>Nivolumab</b>                     | 240 mg (or 3 mg/kg if <40 kg) · or 480 mg                 | q2wk · q4wk        |
| <b>Nivolumab + ipilimumab</b>        | nivo 3 mg/kg + ipi 1 mg/kg × 4, then nivo maintenance     | q3wk → q2–4wk      |
| <b>Aspirin (PIK3CA-mutant)</b>       | 100–162 mg PO daily × 3 years post-op                     | stage II–III       |
| <b>FOLFOX · CAPEOX · mFOLFIRINOX</b> | named only here — dosing on the colon page                | —                  |

## BIOMARKER → THERAPY

| Alteration                                     | Therapy                                                                                                                                     |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>dMMR / MSI-H or POLE-POLD1 (TMB &gt;50)</b> | Neoadjuvant/definitive immunotherapy up to 6 months — <b>preferred over TNT</b> ; dostarlimab, nivolumab or pembrolizumab                   |
| <b>Same, metastatic</b>                        | ICI preferred if no prior exposure: nivolumab ± ipilimumab, pembrolizumab, dostarlimab, cemiplimab, retifanlimab, toripalimab, tislelizumab |
| <b>pMMR/MSS, TMB ≥10</b>                       | Pembrolizumab in the refractory setting (KEYNOTE-158)                                                                                       |
| <b>PIK3CA, stage II–III</b>                    | Aspirin 100–162 mg daily × 3 years after surgery                                                                                            |
| <b>ctDNA</b>                                   | Prognostic only — <b>not</b> for treatment decisions outside a trial                                                                        |

## SURVEILLANCE

- Same as colon cancer for stage I–IV, **except after transanal local excision**
- Transanal local excision:** proctoscopy (with EUS or MRI) **q3–6 months × 2 years**, then q6 months to 5 years total
- Colonoscopy at **1 year**; if advanced adenoma repeat in 1 year, otherwise 3 years then every 5 years
- dMMR on immunotherapy: re-evaluate **q2–3 months**; MSS metastatic being converted to resectable: **q2 months**

## PEARLS

- FOLFOXIRI is not recommended** when resection is contraindicated after neoadjuvant therapy
- Resection is **preferred over ablation or SBRT** for oligometastatic liver- or lung-only disease
- Favorable pT3 N0 (well/moderately differentiated, <2 mm mesorectal invasion, no lymphovascular involvement, upper rectum) may go to surveillance rather than adjuvant therapy
- pT1–3 N1 can be treated with FOLFOX or CAPEOX alone, without radiation